HYPERION AI

SYSTEM DESIGN DOCUMENT

AI CRM Agent

Claude agents that read and write CRM safely - MCP tools, memory, confirmation-gated writes

PREPARED FOR

Firms extending Claude into CRM-integrated chatbot capabilities

Vitalijus Alsauskas

Software and AI Engineer

Hyperion AI - Production AI engineering for founders and agencies

https://hyperionai.dev · 2026-04-19

Production Claude Agents with MCP + CRM — System Design

Prepared for	Firms extending Claude into CRM-aware chatbots
Prepared by	Vitalijus Alsauskas · Hyperion AI
Scope	MCP tools for CRM reads/writes, three-layer memory, confirmation-gated writes, immutable audit log

1. Problem Framing

Claude + CRM chatbots fail in two predictable ways:

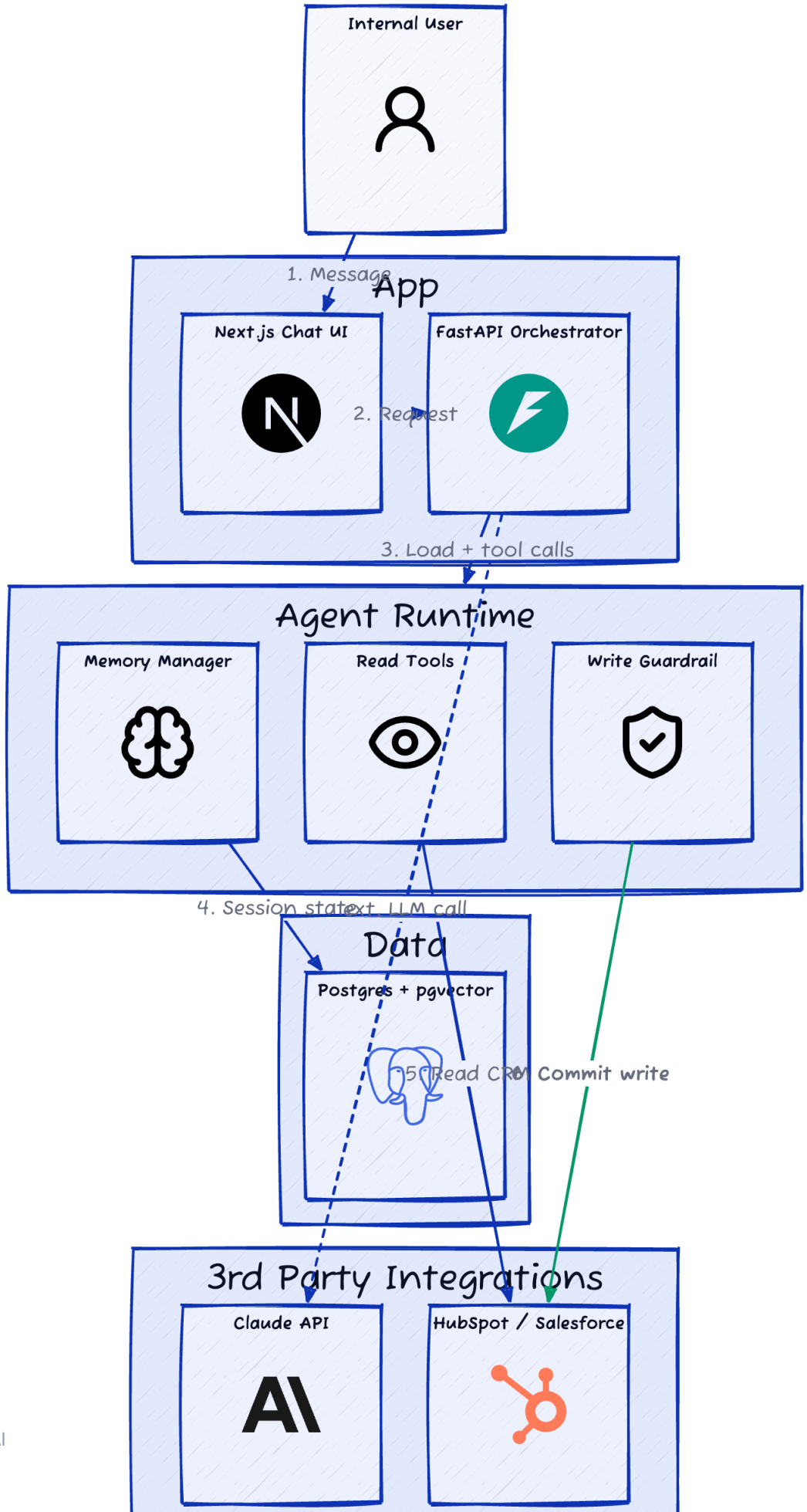
- **Memory gap** — the bot remembers this conversation but forgets everything from last week; users re-explain context every session
- **Silent writes** — the bot mutates CRM records without review; users find surprise entries on Monday morning

The usual responses are both bad:

- **Blind trust** — the agent writes freely, users lose faith after one wrong update
- **Too many guardrails** — the agent can't do anything useful, becomes a toy

This design shows the **pattern that works in production**: MCP tools give the agent a proper interface to your CRM, three layers of memory so users don't repeat themselves, and confirmation-gated writes so the agent drafts but never silently changes records.

2. System Diagram


Internal User
1. Message
App
Next.js Chat UI
2. Request
FastAPI Orchestrator
3. Load + tool calls
Agent Runtime
Memory Manager
Read Tools
Write Guardrail
4. Session state + LLM call
Data
Postgres + pgvector
5. Read CRM
Commit write
3rd Party Integrations
Claude API
HubSpot / Salesforce

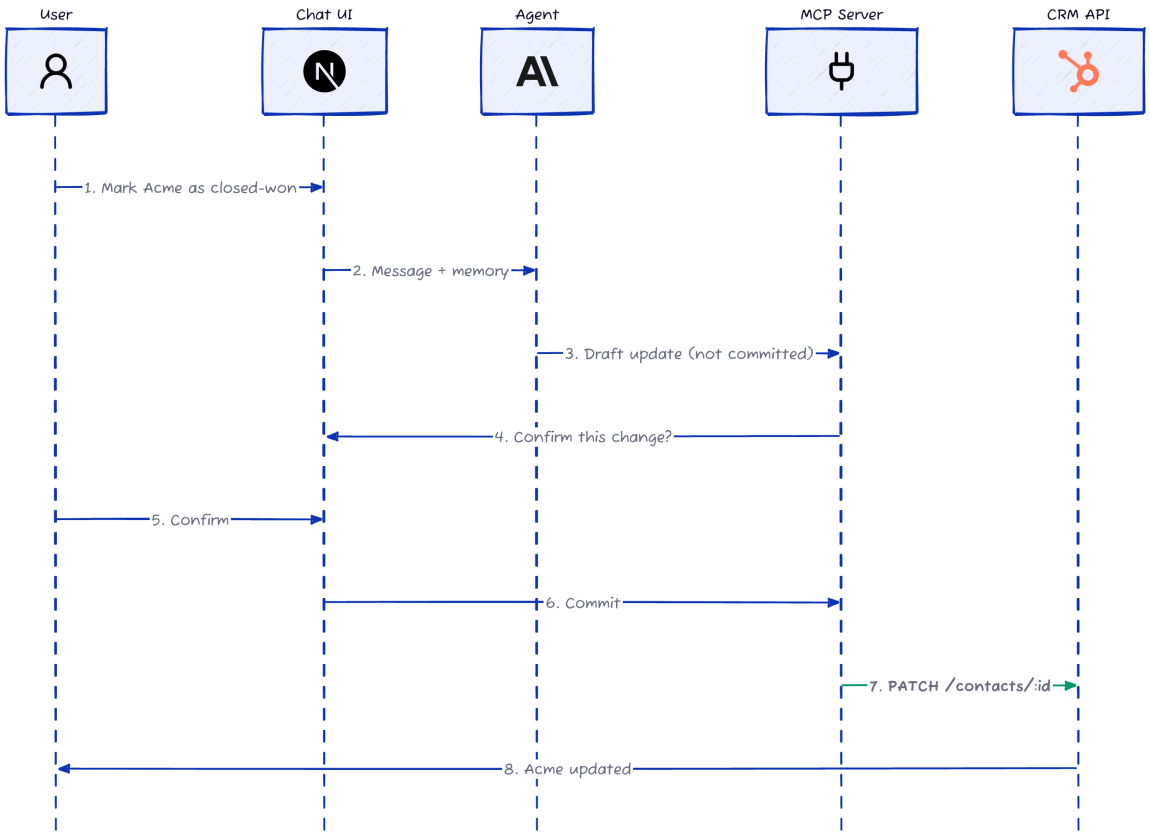
Every CRM capability is exposed as an **MCP tool**. **Read tools** flow directly. **Write tools** route through a guardrail that creates a draft, asks the user to confirm, and only commits on approval. Every tool call is **audit-logged**.

Source: `docs/architecture.d2`

3. Tool Choices

LAYER	PICK	WHY THIS	ALTERNATIVES CONSIDERED
Chat UI	Next.js + React streaming	Server Components for auth, streaming for response UX	Plain React SPA (weaker streaming), Streamlit (not production-ready)
Agent Runtime	FastAPI + Anthropic SDK	Async-native, strong typing, Python ecosystem alignment	Node + `@anthropic-ai/sdk` (fine, Python preferred), LangChain (too much abstraction)
Tool Layer	Model Context Protocol (MCP)	Clean contract, swap CRMs without agent changes, growing standard	Custom function calling (CRM lock-in), LangChain tools (heavier)
Session memory	Postgres rows	Durable, structured, queryable by user + time	Redis only (TTL loss), in-process (restart loss)
Long-term memory	pgvector	Semantic recall of past conversations, co-located with Postgres	Pinecone (extra service), Weaviate (overkill)
Audit log	Immutable S3 + Postgres mirror	Append-only, compliance-grade, queryable	Regular DB table (mutable), CloudTrail only (misses app context)
CRM	HubSpot / Salesforce / Pipedrive APIs	Official, documented, rate-limited in known ways	Scraping (fragile), Zapier middleware (extra hop)

4. Data Flow



Read flow

1. **User message** hits the chat UI, streams to agent
2. **Memory load** — three layers: session (last N turns), user (preferences), account (company-wide)
3. Agent sends **prompt + MCP tool schema** to Claude
4. **Claude picks tools** — can chain multiple tool calls
5. Each tool call hits **MCP server** → **CRM API** → **audit log**
6. **Claude composes final answer**; agent streams back to UI
7. **Turn persists** to Postgres + facts embed into vector store

Write flow (confirmation-gated)

1. User asks for a mutating action ("*mark Acme as closed-won*")
2. Claude returns a tool call for `update_stage`
3. **Agent detects mutation**, routes through guardrail
4. Guardrail creates a **draft row** in Postgres
5. Agent returns **"Draft: change Acme stage to closed-won. Confirm?"** to UI
6. User clicks confirm → UI sends commit request

7. Agent fires **actual CRM write**, logs commit to audit with before/after
8. User sees confirmation message

Error paths

FAILURE	HANDLING
Claude timeout	Exponential backoff; fallback to Claude Haiku after 2 failures
MCP tool error	Bubbled up as tool result with error context; agent retries or informs user
CRM API error	Tool result = error; agent retries or says "*couldn't reach the CRM, try again*"
User rejects draft	Draft row marked `rejected`; nothing commits to CRM
Memory load failure	Fall back to empty memory; log warning; conversation proceeds with lower quality

---

5. Component Breakdown

COMPONENT	STACK	RESPONSIBILITY
Chat UI	Next.js 15 + Server Components + streaming SSE	Message input, streaming output, confirmation drawer, per-user audit view
Agent Orchestrator	FastAPI + Anthropic SDK	Tool-use loop, prompt composition, memory integration
Memory Manager	Python + SQLAlchemy + pgvector	Three-layer memory retrieval + persistence
MCP Server	Python MCP SDK	Tool registration, per-backend CRM adapter (`hubspot.py`, `salesforce.py`, `pipedrive.py`)
Write Guardrails	Agent middleware	Intercept mutating tool calls, draft persistence, confirmation flow
Audit Logger	FastAPI middleware + S3 writer	Every tool call + commit logged with full context
Admin UI	Next.js `/admin` route	View audit trail, search conversations, revoke user access

6. Implementation Phases

PHASE	WEEKS	SHIPS	KEY MILESTONES
1. Foundation	1-2	Read-only chatbot answering CRM questions with session memory	Chat UI, agent orchestrator, 3 MCP read tools, session memory in Postgres, one CRM backend, audit log
2. Memory + Writes	3-4	Three-layer memory + confirmation-gated writes	User + account memory, pgvector, draft persistence, confirmation drawer UI, one write tool (`log_activity`), admin UI
3. Production	5-6	Second CRM backend + observability + deploy	Additional CRM adapter if needed, Sentry wired in, runbook, knowledge transfer session

7. Risks & Mitigations

#	RISK	MITIGATION
1	**Agent writes to CRM silently** — surprise records	Confirmation-gated flow. Every mutating tool goes through guardrail. No commit without explicit user approval. Drafts persist in DB so nothing lost between draft and commit.
2	**Agent hallucinates CRM data** instead of calling the tool	System prompt forbids answering CRM questions without a tool call. Strict tool schemas. Audit log surfaces any hallucinations in review.
3	**PII in conversation history** violates compliance	Anthropic zero-retention flag. Memory redacts sensitive fields before embedding. User-level data isolation (no cross-user memory access).
4	**CRM API rate limits** during heavy usage	MCP server caches recent reads (60s TTL), batches where possible, exponential backoff on 429. Hard per-user concurrency limit.
5	**Audit log tampering or loss**	S3 Object Lock for immutability. Postgres mirror for fast query. Cannot delete or modify existing audit rows via app code.

8. What I Need From You

To move into Phase 1, a few inputs from your side:

- **Which CRM** — HubSpot, Salesforce, Pipedrive, or other?
- **Top 5 reads + top 2 writes** the chatbot should cover in Phase 1
- **3-5 sample user scenarios** — example conversations you want the bot to handle
- **Current Claude setup** — using the API directly or via a vendor wrapper? Any existing MCP work?
- **Compliance requirements** — data residency, retention, redaction rules?
- **30-minute kickoff call** — to walk through this doc against your actual setup

This document is yours to keep regardless of whether we end up working together. The MCP + confirmation pattern here will outlive any specific CRM integration.